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# GEN AI + Agentic AI with Python Developer Program
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**Trainer:** Mr. Ashok
**IT Experience:** 12+ Years
**Training Experience:** 10+ Years
**Ashok IT Started:** 2020
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Program Overview

Build your career in the fast-growing world of **Generative AI, Agentic AI, Python, LLMs, RAG Systems, AI Agents, LangChain, LangGraph, MCP, and MLOps**.

This program is designed to take learners from **basic level** to advanced project development, even if they do not have prior programming experience.

Our goal is simple:

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**Zero to Hero in Gen AI + Agentic AI with Python**
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Pre-Requisites

No pre-requisites required.

This program is suitable for beginners. We will start from the basics and gradually move towards advanced real-time project development.

You will learn:

- * Python programming from scratch
- * AI and ML fundamentals
- * Generative AI application development
- * RAG-based systems using Vector DB
- * AI agents development
- * Agentic workflows
- * Real-time project implementation
- * Deployment and MLOps basics
- * Interview preparation and resume building

Course Content

Module 1: Python Programming

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**Core Python + Advanced Python**
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In this module, students will learn Python from the basics.

Topics Covered

- * Introduction to Python

- * Python installation and environment setup
- * Variables and data types
- * Operators
- * Conditional statements
- * Loops
- * Strings
- * Lists, tuples, sets, and dictionaries
- * Functions
- * Modules and packages
- * File handling
- * Exception handling
- * Object-Oriented Programming (OOPS)
- * Advanced Python concepts
- * Working with APIs
- * Database connectivity
- * Real-time Python coding practices

Module 2: Python Libraries for AI Development

This module helps students understand the important Python libraries used in AI, ML, Data Science, and Gen AI projects.

Topics Covered

NumPy

- * Arrays
- * Matrix operations
- * Numerical computations
- * Data processing basics

Pandas

- * Series and DataFrames
- * Data cleaning
- * Data filtering
- * Data transformation
- * CSV and Excel file handling

Matplotlib

- * Data visualization
- * Line charts
- * Bar charts
- * Pie charts
- * Scatter plots
- * Visual analysis

Probability and Statistics

- * Mean, median, mode
- * Variance and standard deviation
- * Probability basics
- * Distributions
- * Correlation
- * Hypothesis testing basics

Scikit-Learn

- * ML model development

- * Data preprocessing
- * Model training
- * Model testing
- * Accuracy evaluation

TensorFlow

- * Deep learning basics
- * Neural networks
- * Model building
- * Training deep learning models

FastAPI

- * REST API development
- * Building AI APIs
- * Connecting ML/Gen AI models with web applications
- * API testing

Module 3: Machine Learning and Deep Learning Algorithms

This module focuses on ML and DL concepts required for AI developers.

Machine Learning Topics

- * Introduction to Machine Learning
- * Types of Machine Learning
 - * Supervised Learning
 - * Unsupervised Learning
 - * Reinforcement Learning
- * Regression algorithms
- * Classification algorithms
- * Clustering algorithms
- * Decision Trees
- * Random Forest
- * KNN
- * SVM
- * Naive Bayes
- * Model training and testing
- * Model evaluation metrics
- * Real-time ML project implementation

Deep Learning Topics

- * Introduction to Deep Learning
- * Artificial Neural Networks
- * Activation functions
- * Backpropagation
- * Optimizers
- * CNN basics
- * RNN basics
- * Transformers introduction
- * Deep learning model building

Module 4: LLMs and Prompt Engineering

This module helps students understand Large Language Models and how to work with popular AI platforms.

Topics Covered

- * Introduction to LLMs
- * How ChatGPT works
- * Tokens and context window
- * Temperature and model parameters
- * Prompt engineering basics
- * Advanced prompt engineering
- * Zero-shot prompting
- * Few-shot prompting
- * Chain-of-thought style prompting
- * Role-based prompting
- * Prompt templates
- * OpenAI API integration
- * Google Gemini API integration
- * Claude API integration
- * Building AI-powered applications using APIs
- * Best practices for prompt design

Module 5: Generative AI Application Development and RAG Systems

This module focuses on building real-world Gen AI applications.

Topics Covered

- * Introduction to Generative AI
- * Text generation applications
- * Chatbot development
- * Document Q&A systems
- * PDF-based question answering
- * Embeddings
- * Vector databases
- * Semantic search
- * RAG architecture
- * Chunking strategies
- * Document loading
- * Vector storage
- * Retrieval techniques
- * Building RAG applications
- * RAG with PDFs, websites, and databases
- * AI chatbot with knowledge base
- * Real-time Gen AI project development

Module 6: Agentic AI, LangChain, LangGraph, MCP and AI Agents

This is one of the most important modules of the program. Students will learn how to build intelligent AI agents that can plan, reason, use tools, and complete tasks.

Topics Covered

Agentic AI

- * What is Agentic AI?
- * Difference between Gen AI and Agentic AI
- * AI agents architecture

- * Agent workflows
- * Planning and reasoning
- * Tool usage by agents
- * Multi-step task execution

LangChain

- * Introduction to LangChain
- * LLM integration
- * Prompt templates
- * Chains
- * Memory
- * Tools
- * Agents
- * RAG using LangChain
- * LangChain project implementation

LangGraph

- * Introduction to LangGraph
- * Graph-based AI workflows
- * Nodes and edges
- * State management
- * Conditional workflows
- * Multi-agent workflows
- * Human-in-the-loop workflows
- * LangGraph project implementation

MCP

- * What is Model Context Protocol?
- * Why MCP is useful
- * Connecting AI agents with external tools
- * MCP server and client basics
- * Real-time MCP use cases

AI Agents Development

- * Building custom AI agents
- * Tool-calling agents
- * Web search agents
- * Database agents
- * File analysis agents
- * Multi-agent systems
- * Agentic AI real-time projects

Module 7: MLOps + LLMOps

This module helps students understand how to deploy and maintain AI applications in production.

Topics Covered

- * Git and GitHub
- * Version control
- * Docker basics
- * Dockerizing Python applications
- * Kubernetes basics
- * CI/CD pipeline overview
- * Cloud deployment basics
- * Model deployment

- * API deployment
- * Monitoring AI applications
- * Logging and exception handling
- * Production-ready project structure

Module 8: Interview Preparation and Resume Building

This module prepares students for job opportunities in AI, ML, Gen AI, and Agentic AI roles.

Topics Covered

- * Python interview questions
- * ML interview questions
- * Deep learning interview questions
- * Gen AI interview questions
- * LLM interview questions
- * RAG interview questions
- * LangChain and LangGraph interview questions
- * AI agents interview questions
- * Project explanation preparation
- * Resume building
- * GitHub profile preparation
- * LinkedIn profile guidance
- * Mock interview guidance
- * Job role preparation

Who Can Attend?

This program is suitable for:

1. Freshers
2. Developers
3. Operations Engineers
4. Testers
5. Non-IT Students
6. Final-year students
7. Working professionals who want to move into AI
8. Career gap candidates
9. People who want to learn AI application development from scratch

Why This Course Is Important Today

Earlier, one of the most difficult roles in IT was the **development role**, because developers had to write most of the code manually.

Today, the software industry is changing because of AI tools.

Earlier:

Manual Coding Process :

Developers had to write almost everything on their own.

Today:

AI-Assisted Development / Vibe Coding :

Developers explain the requirement clearly to AI tools, and AI helps in generating code, improving productivity, debugging, testing, and documentation.

This means development is becoming easier for people who understand:

- * Requirements
- * Logic
- * AI tools
- * Prompting
- * Application architecture
- * Testing
- * Deployment

That is why learning **Python + Gen AI + Agentic AI** is a powerful career opportunity.

Course Details

****Start Date:**** Today
****Class Timings:**** 7:30 AM to 9:00 AM IST
****Days:**** Monday to Saturday
****Duration:**** 3 to 4 Months
****Mode:**** Online Live Training
****Trainer:**** Mr. Ashok
****Course Fee:**** ₹1,25,000

What Students Will Get

- * Live online training
- * Soft copy materials
- * Class recordings
- * Recordings access for 1 year after course completion
- * Real-time project development
- * Interview preparation
- * Resume building support
- * Practical coding sessions
- * AI tools exposure
- * Gen AI and Agentic AI project knowledge

Job Roles After This Program

After completing this program, students can prepare for roles such as:

1. Data Scientist
2. AI Engineer
3. Machine Learning Engineer
4. Gen AI Developer
5. Agentic AI Developer
6. AI Application Developer
7. AI Tester
8. Python Developer
9. LLM Application Developer
10. RAG Developer
11. AI Automation Developer

Final Message

Join the ****GEN AI + Agentic AI with Python Developer Program**** and start building your career in the next generation of software development.

Learn Python, AI, ML, Deep Learning, LLMs, Prompt Engineering, RAG, LangChain, LangGraph, MCP, AI Agents, and MLOps from basic to advanced level.

****Start your AI career journey with Ashok IT.****

What is Data Science?

Data is the starting point of Data Science.

Using data, we can create intelligence. Using intelligence, we can build systems that can:

- * Learn from data
- * Understand human language
- * Generate content
- * Make predictions
- * Take actions

Definition

****Data Science**** is the process of collecting, cleaning, analyzing, and understanding data to make smart decisions.

In simple words:

****Data Science means extracting useful information from data.****

Data Science Life Cycle

The Data Science life cycle contains the following steps:

1. ****Collect Data****
Gather data from different sources such as files, databases, websites, sensors, applications, and APIs.
2. ****Clean Data****
Remove duplicate data, missing values, incorrect values, and unwanted information.
3. ****Analyze Data****
Study the data to understand patterns, trends, and relationships.
4. ****Visualize Data****
Represent data using charts, graphs, dashboards, and reports.
5. ****Build Model****
Create a machine learning model based on the problem requirement.
6. ****Train Model****
Train the model using historical data so that it can learn patterns.
7. ****Test Model****
Check whether the model is giving correct predictions or not.
8. ****Deploy Model****

Make the model available for real-time usage in applications.

9. ****Monitor Results****

Continuously observe the model performance and improve it when required.

What is Artificial Intelligence?

Artificial Intelligence is a branch of Data Science and Computer Science.

AI aims to create machines or systems that can think and act like humans.

AI is used to create intelligent machines and intelligent software systems.

Simple Definition

****AI means making computers perform tasks that normally require human intelligence.****

In normal programming, we tell the computer exactly what to do.

But in Artificial Intelligence:

1. The machine observes data
2. The machine learns from data patterns
3. The machine improves over time
4. The machine makes decisions on its own

AI Real-Life Examples

Artificial Intelligence is used in many real-time applications, such as:

1. Self-driving cars
2. Fraud detection systems in banks
3. Spam detection in emails
4. ChatGPT answering questions and generating images
5. GitHub Copilot generating code
6. Voice assistants like Siri, Alexa, and Google Assistant
7. Product recommendations in Amazon and Flipkart
8. Movie recommendations in Netflix and YouTube

Why is AI Important Today?

Artificial Intelligence is very important today because it helps us to:

1. Automate repetitive tasks
2. Reduce human errors
3. Save time and cost
4. Improve business decision-making
5. Work faster and more accurately
6. Build smart applications for different industries

AI is used in almost every industry, such as:

- * Medicine
- * Farming
- * Finance
- * Education

- * E-commerce
- * Banking
- * Software development
- * Transportation

What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence.

Machine Learning allows computers to learn patterns from data without writing programs explicitly for every condition.

Simple Definition

****Machine Learning is a technique where the computer learns from data and makes predictions or decisions.****

In Machine Learning:

1. We give a large amount of data to the computer
2. The computer finds patterns in that data
3. The computer uses those patterns to make predictions or decisions

Important Note

Machine Learning does not follow only fixed rules like normal programming.

Instead, it learns from examples and data.

Example 1: Cat and Dog Image Classification

Suppose we show a machine 1000 pictures of cats and dogs.

The Machine Learning model learns the patterns of cats and dogs.

After training, if we give a new image, the model can identify whether it is a cat or a dog.

Example 2: Email Spam Filter

In an email spam filter:

1. The Machine Learning model learns from thousands of emails
2. It observes which emails are marked as spam and which emails are safe
3. Later, it automatically identifies spam emails in your inbox

Daily Life Examples of Machine Learning

Machine Learning is used in many daily life applications:

1. ****Netflix / YouTube Recommendations****
Learns your watching patterns and suggests similar videos.
2. ****Google Photos****
Recognizes faces, pets, objects, and places.

3. **Online Shopping**
Amazon and Flipkart show products based on your past purchases and search history.
4. **Credit Card Fraud Detection**
Detects unusual transactions and alerts the bank or customer.
5. **Google Maps**
Predicts traffic and suggests the fastest route.

How Machine Learning Works in Projects

Machine Learning projects generally follow these steps:

Step 1: Collect Data

Collect data from different sources such as images, text, numbers, articles, websites, databases, and APIs.

Step 2: Train the Model

Train the model using data. During training, the model learns patterns from the data.

Step 3: Test the Model

Test the model to check whether it is giving correct predictions or not.

Step 4: Use the Model

Use the trained model to make predictions on new unseen data.

Summary

Machine Learning = Learn from data → Improve over time → Make predictions

What is Deep Learning?

Deep Learning is a subset of Machine Learning.

Deep Learning uses artificial neural networks to learn from large amounts of data.

Deep Learning algorithms are inspired by the human brain.

Deep Learning is part of Machine Learning, but it is more powerful for complex problems.

Machine Learning vs Deep Learning

Machine Learning

- * Machine Learning is a subset of AI
- * ML models learn from data using algorithms
- * Works well with small to medium data
- * Mostly works with structured data
- * Training time is usually faster
- * Easier to understand and explain

Deep Learning

- * Deep Learning is a subset of Machine Learning
- * DL uses neural networks with many layers
- * Needs a huge amount of data
- * Works well with unstructured data such as images, audio, video, and text
- * Training time is usually slower
- * Gives better accuracy for complex problems

Where is Deep Learning Used?

Deep Learning is used in many advanced applications, such as:

1. **Face Recognition**
Detects and recognizes faces using deep neural networks.
2. **Speech to Text**
Converts spoken words into text.
3. **Text to Speech**
Converts written text into voice.
4. **Self-Driving Cars**
Detects traffic signals, road signs, pedestrians, and vehicles.
5. **Medical Diagnosis**
Detects diseases from X-rays, MRI scans, and medical images.
6. **Chatbots and AI Assistants**
Understand user questions and generate human-like responses.

When to Use Machine Learning?

Machine Learning is suitable when:

1. Data is limited
2. Problem is simple
3. Model should be easy to explain
4. High accuracy is not extremely critical
5. Data is mostly structured

Examples:

- * Loan approval prediction
- * Customer churn prediction
- * Sales forecasting
- * Email spam detection

When to Use Deep Learning?

Deep Learning is suitable when:

1. Large amount of data is available
2. Data is in the form of images, audio, video, or text
3. Problem is complex
4. High accuracy is very important

5. Powerful hardware like GPU is available

Examples:

- * Face recognition
- * Voice assistants
- * Image classification
- * Self-driving cars
- * Generative AI applications

What is NLP?

NLP stands for Natural Language Processing.

Natural Language Processing is a branch of Artificial Intelligence that helps computers understand, interpret, and generate human language.

Simple Definition

****NLP means teaching computers to understand English, Telugu, Hindi, or any other human language.****

Humans communicate using language, sentences, emotions, tone, and grammar.

But computers understand numbers, mainly 0s and 1s.

NLP acts as a bridge between humans and computers.

What Can NLP Do?

Using NLP, computers can perform the following activities:

1. Read text
2. Listen to speech
3. Understand meaning
4. Analyze human emotions
5. Generate answers
6. Translate languages
7. Summarize large documents
8. Extract important information from text

NLP Real-Life Examples

ChatGPT and Gemini

ChatGPT and Gemini understand user questions and generate human-like answers.

Google Translate

Google Translate converts one language into another language, such as Telugu to English or Hindi to Tamil.

Siri and Alexa

Siri and Alexa understand spoken commands and respond to users.

Email Spam Detection

NLP helps identify whether an email is spam or not.

Sentiment Analysis

Companies use NLP to understand whether customer feedback is positive, negative, or neutral.

Final Summary

Data Science, AI, ML, DL, and NLP are connected to each other.

- * **Data Science** extracts useful information from data.
- * **Artificial Intelligence** makes machines intelligent.
- * **Machine Learning** allows machines to learn from data.
- * **Deep Learning** uses neural networks to solve complex problems.
- * **NLP** helps computers understand and generate human language.

In simple words:

Data → Data Science → AI → ML → DL → NLP / Gen AI Applications

These technologies are the foundation for modern intelligent applications like ChatGPT, Gemini, self-driving cars, recommendation systems, fraud detection systems, and AI-powered software tools.

===== ## What is Generative AI? =====

Generative AI is a type of AI that can create new content.

It can generate

- 1) text
- 2) code
- 3) images
- 4) audio
- 5) video
- 6) report
- 7) email template

=> GEN AI models don't just analyze the data, they generate something new...

Traditional AI focuses on:

- Classification (spam or not spam)
- Predication (price, weather)
- Detection (objects, fraud)

But Generative AI goes beyond that.

=> It learns patterns from huge amounts of data and then uses those patterns to generate original content that looks like it was created by human.

GEN AI Examples

- Chat GPT : Creates text

- MidJourney/DALL-E : Creates images
- Suno : Creates music
- RunWayML : Creates videos
- GitHub Copilot : Writes code
- Gamma : Create PPT

=> If you train GEN AI model with 1000 songs, then it can create new song which is not a copy of any single song.

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What is Agentic AI?
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=> Agentic AI is a type of Artificial Intelligence that can think, decide, and take actions on its own to achieve a goal

=> Agentic AI is used to develop intelligent AI assistant or AI agents

=> Traditional AI can answer only our questions where as Agentic AI can "plan + Act + Complete Task independently".

~~ Gen AI : Only generates content + Needs user prompt + Static response

~~ Agentic AI : Takes actions to achieve goals + Can work autonomously + Interactive, decision-making

Note: Agentic AI behaves like a digital worker.

Agentic AI Examples

- An AI that books your flight tickets after comparing prices
- AI that reads your emails and drafts replies
- AI DevOps agents that deploy apps automatically
- AI customer support agents that solve issues end-to-end
- AI personal assistant that schedules meetings

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Summary
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** Data Science --> Find useful insights from data to make smart business decisions

** AI =>> Big Umbrella, it is used to make systems more intelligent.

** ML =>> System learns from existing data and make decions/predictions on new data

** DL => ML that uses neural networks to find patterns on complex data

** NLP => AI that understands human language

** GEN AI => AI that creates new content

** Agentic AI => AI that takes actions (AI Agent)